Deployment Recommendation Report

Course: INFT 41000, Deploying Artificial Intelligence Systems
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Use Case Title: Predictive Maintenance for a Manufacturing Plant
GitHub Commit Link:

AI-Use Statement

In compliance with the course academic integrity policy, generative AI (Google Gemini) was used as a supplementary tool for brainstorming potential solution risks, theoretical explanations of cloud edge patterns, and proofreading or grammar suggestions. All the core analysis, readiness scoring, framework assessments, and final architectural recommendations were produced independently. All the numeric cost calculations, assumptions, service names, and URLs were manually verified against the official Microsoft Azure technical documentation and pricing pages to ensure accuracy and avoid AI-generated inaccuracies.

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1. Use Case and Readiness Snapshot

1.1 Use Case Description

The proposed AI system is intended for industrial predictive maintenance. Its purpose is to predict possible malfunction of a manufacturing plant's equipment based on continuous telemetry data streams from high-frequency vibration, thermal, and acoustic sensors attached to the machinery. It will be used by reliability engineers and machinery maintenance personnel to receive predictions of upcoming malfunctions and preemptively shut down the equipment for maintenance. The model will be trained on the data retrieved from the plant's telemetry equipment and the maintenance logs. However, the most critical stage is the constant functioning in production conditions and responding accordingly. The system should be able to process incoming data streams and perform localized inference even when the factory's network is unavailable.

1.2 Production AI Requirement

This system has to be a production AI and not an experimental one because a plant downtime in production has losses in tens of thousands of dollars per hour, while a mechanical seizure that went undetected by the system would create significant safety risks for floor operators. This requires 99.9% up-time, fail-safes in the inference process, and immediate response time, which a simple notebook experiment or a one-off batch script cannot provide.

1.3 Readiness Scores

The system's production readiness across the six core dimensions was evaluated using the Week 2 framework. The table below outlines each dimension, specific evidence from the manufacturing plant floor, and the assigned score:

Dimension	Evidence	Score
Data	Historical sensor logs are available for the primary machines, but vibration telemetry labeling across plant shifts needs to be reviewed and standardized.	3
Model	Time-series classification and anomaly detection Proof of Concept models demonstrate accuracy of over 88% on clean historical validation sets.	4
Infrastructure	The factory floor is populated with legacy PLCs and lacks inference-grade compute gateways, but an enterprise-	2

	level Azure cloud tenant is already provisioned.	
Organizational	The company has a competent workforce of reliability engineers and mechanical technicians, but the IT department is understaffed and lacks any dedicated MLOps personnel.	3
Risk	While false alarms do not present a direct safety hazard, but the cost of missed breakdowns would be prohibitively expensive.	4
Cost	The ongoing operational expenditure costs of cloud storage for telemetry data are authorized, but the capital expenditure spend on edge gateways is tightly controlled.	3

1.4 Effect of Lowest-Scoring Dimensions

The two lowest scoring areas are the Infrastructure and the Organizational. Weak infrastructure means that the model inferencing cannot rely solely on cloud due to unstable uplink bandwidth from the factory, which requires having edge industrial gateways on the factory floor. Weak organizational capacity heavily influences the architecture towards adopting managed cloud platform services for centralized model training, continuous model monitoring, and automated model deployment pipelines instead of utilizing bare-metal servers.

2. Deployment Environment Evaluation

2.1 Environment Fit Scorecard

The four environment fit framework questions were evaluated in order to determine architectural placement between on-premise edge and cloud:

Question	Score	Evidence
Data sensitivity	3	Proprietary industrial machine telemetry and operational logs; trade secret data, but free from consumer PII or regulatory residency laws.

Cost pattern	3	Aside from monthly edge compute, sensor telemetry streaming is steady, but GPU retraining costs spike intermittently during monthly model updates.
Operational capacity	5	A small plant IT team with no MLOps designated personnel can't maintain, patch, and cool on-premise GPU server clusters 24/7.
Latency	1	On the factory floor, real-time alerts need to be delivered within a sub-second window; network outages shouldn't interrupt machine failure detection.

Apply decision rules to this scenario: Hard Limiting: Only Latency hit 1 which forces the inference workload to run on-premise at the factory edge. Spread: Highest minus the lowest: 5 – 1 = 4 (Spread >= 2) this strictly demands Hybrid over pure cloud or pure on-premise.

2.2 Environment Comparison

Comparison of the three environments for this manufacturing use-case scenario:

• **On-Premises:** highly controlled data, no WAN latency risks but also means huge capital expenditures (CapEx) into redundant GPUs and factory IT administrative overhead

• **Cloud:** extremely elastic GPU burst capacity and no maintenance for the compute hardware but at an intolerable risk for production monitoring downtime in case of plant internet outages

• **Hybrid**: local edge runtimes for low-latency or offline inference plus cloud for intermittent model retraining with model registry and long-term telemetry storage aggregation

2.3 Recommendation and Justification

My Recommendation: Hybrid (Edge + Cloud) architecture

Justification: Decision driven by the Latency and Cost Pattern framework questions. Latency or Network considerations require that machine inference must occur on-premise in a factory floor containerized edge runtime, since any network outage for an external party would lead to catastrophic loss of situational awareness around imminent machine failures. On the other hand, the Cost Pattern necessitates that intensive model retraining and historical telemetry aggregation must occur in the public cloud to avoid large capital expenditures on on-premise

GPU clusters that would be largely unused between model retraining cycles. These two factors together are sufficient for a hybrid decision according to the scorecard decision rule.

2.4 Three-Year Cost Comparison

Assumption	Value	Basis
Monitored machines	50 production units	High-priority rotating equipment across the plant floor requiring continuous vibration and thermal monitoring.
Sensor data ingestion	~5 TB raw / month	50 machines streaming 24/7 high-frequency telemetry; processed and compressed locally before cloud sync.
Edge inference nodes	50 industrial gateways	Ruggedized IoT gateways mounted at machinery cells to provide offline, sub-second inference.
Monthly retraining compute	50 GPU hours / month	Scheduled monthly batch retraining and hyperparameter tuning on updated plant telemetry.
Staff time, hybrid	0.2 FTE	Monitoring pipelines, reviewing drift alerts, and periodic model deployment via managed PaaS.
Staff time, on-prem	0.5 FTE	Patching OS/drivers, physical hardware maintenance, disk management, and manual ML model updates.
Loaded staff cost	$110,000 CAD per FTE per year	Ontario technical engineering market salary data for system or IT support.
Exchange rate	1 USD = 1.37 CAD	Standard bank exchange rate as of October 2026.

Cloud line item	Rough monthly (USD)	How to price it
Azure Machine Learning (GPU compute)	~$95	~50 hours/month on Standard NC6s_v3 GPU instances for monthly model retraining runs.
Azure IoT Hub (Standard Tier S2)	~$50	1 unit handling bi-directional telemetry streaming and device twins for 50 edge gateways.
Azure Data Lake Gen2 & Blob Storage	~$45	~5 TB rolling telemetry archive with lifecycle tiering to Cool/Archive storage.
Azure Container Registry &	~$25	Private container registry for

Key Vault		edge model images, secrets, and mTLS certificates.
Azure Monitor & Log Analytics	~$40	Ingesting diagnostic telemetry, health metrics, and edge gateway heartbeat logs (~15 GB/mo).
Virtual Network & Private Endpoints	~$30	Secure private connectivity and data transfer between IoT Hub and cloud workspace.
Total	**~$285 USD (~$390 CAD) per month**	**Baseline monthly operational cloud spend.**

Cost item (CAD)	**Hybrid (Recommended), 3 years**	**On-prem, 3 years**
Central GPU server for training & failover	$0	$80,000 (Dual-GPU server rack, SAN storage, redundancy)
Factory edge gateways ($500 ea)	$25,000	$25,000 (50 industrial edge nodes required in both)
Cloud services	$18,500 (~$390/mo plus 30% spike buffer)	$0
Power and cooling (server room 24/7)	$0	$18,000
Hardware support & maintenance contract	$0	$12,000
Staff time	$66,000 (0.2 FTE over 3 years)	$165,000 (0.5 FTE over 3 years)
One-time initial setup & integration	$15,000	$25,000
Total	**≈ $124,500 CAD**	**≈ $325,000 CAD**

As of three years, the recommended Hybrid architecture costs less than 40% of a fully On-Premise deployment. Staff time is the most important and variable aspect, and cloud handles model retraining and orchestration for a reasonable price with Microsoft support if required. Factory edge gateways are retained in both options to guarantee offline alert survival.

2.5 Tipping Point

The tipping point that would change this recommendation to a pure cloud architecture is the commissioning of a redundant, carrier-grade Private 5G industrial network coupled with AWS Wavelength / Azure Edge Zones providing guaranteed 99.999% SLA connectivity with sub-5ms cloud latency. This would eliminate the necessity of on-site edge compute.

3. Cloud Platform and Service Selection

3.1 Platform Choice and Trade-offs

Selected Platform: Microsoft Azure.

Justification: The choice of Microsoft Azure was guided by three key factors: (1) Excellent industrial IoT edge support via Azure IoT Edge that delivers premier-edge-to-cloud container deployments; (2) Excellent enterprise connectivity to leverage existing Active Directory and Windows SCADA infrastructure; and (3) Strong MLOps tooling in Azure Machine Learning that natively supports containerization and deployment to edge targets.

Alternatives: AWS was ruled out because its IoT Greengrass platform presents higher deployment friction with the plant's existing Microsoft enterprise stack. Google Cloud was ruled out because it formally deprecated Google Cloud IoT Core, making it ill-suited for deep industrial IoT fleet orchestration.

3.2 Service Map Overview

The architecture maps directly to Microsoft Azure services at each layer:

• **Training Compute:** Azure Machine Learning Serverless Compute / AKS GPU cluster (PaaS/IaaS) for elastic model training

• **Serving or Hosting:** Azure IoT Edge (PaaS runtime on local gateways) with containerized inference

• **Data Storage:** Azure Data Lake Storage Gen2 (PaaS) for archival of historical telemetry/model artifacts

• **Managed AI/ML:** Azure Machine Learning Workspace (PaaS) for MLOps/registry/experiments

• **Networking/Ingestion:** Azure IoT Hub (PaaS) for MQTT/AMQP bi-directional secure messaging between factory gateways and cloud

3.3 Containers vs. Serverless

Justification: (Traffic Pattern -> Execution Duration -> Portability)

The inferencing workload has continuous sensor telemetry as input that is executed 24/7 with strict offline portability requirements. Serverless functions (Azure Functions) have cold start latency and cannot guarantee execution in case of internet outages. Therefore we select containerized microservices (docker containers orchestrated by Azure IoT Edge daemon) for inference. For the cloud retraining workload, select serverless compute (**Azure Machine Learning Serverless Compute**) because these are short-running, intermittent GPU batch workloads.

3.4 Build vs. Buy Strategy

Decision: Buy (Managed PaaS via Azure Machine Learning)

Comparing alternatives: Building a stack of open-source tools (Kubernetes + MLflow + Kubeflow) versus buying Azure ML. The build option provides more customization but requires significant DevOps expertise and takes 6-9 months to implement. Buying Azure ML reduces time-to-value (under 6 weeks) and provides out-of-the-box capabilities like model registry and edge container manifests while minimizing operational complexity.

3.5 Cloud Adoption Framework (CAF)

• **Plan:** Establish business drivers (reduce unplanned downtime by 30%), reference architecture, and cloud governance model

• **Ready:** Configure Azure Landing Zone with Hub-Spoke virtual networks, Role-Based Access Control, and Azure Policy

• **Adopt:** Follow phased rollout. Start with 1 production line then scale to 50-machine plant floor

• **Govern and Operate:** Configure Azure Cost Management budgets with automatic limits on GPU clusters, and Azure Monitor or Defender for IoT security

3.6 Lock-in Assessment and Mitigations

The greatest risk of lock-in would be with Azure IoT Hub and the Azure IoT Edge daemon, as these provide proprietary device connection and deployment mechanisms. Our mitigation for this risk is to ensure all sensor ingestion follows standard MQTT/REST protocols, and that models are implemented as portable docker containers with ONNX model files. This allows us to migrate the inferencing container to AWS IoT Greengrass or K3s if needed.

4. Risks and Decision

4.1 Top Three Deployment Risks

1. Edge Network Disconnect & Telemetry Loss (Technical Risk):

• **Likelihood:** High, **Impact:** Medium

• Solution: Configure Azure IoT Edge to implement local SQLite/disk buffering for 72 hours of sensor telemetry in case of WAN disconnection, with automatic sync when connection is re-established

2. Edge Device Perimeter Breach (Security Shared Responsibility Risk):

• **Likelihood:** Medium, **Impact:** High

• **Solution:** Enable Hardware Security Modules on edge gateways, implement mutual TLS authentication for all sensor data, and place edge nodes on isolated industrial VLANs

3. Uncontrolled Cloud GPU Retraining Costs (Financial Risk):

• **Likelihood:** High, **Impact:** Medium

• **Solution:** Implement hard auto-shutdown timers (2+ hour max) on Azure ML compute targets, utilize low-priority spot instances for experimental training, and set up budget alerts

4.2 Go / Delay / No-Go Decision

Decision: GO (Ship to Pilot)

The project is approved to proceed, conditional on the following three pre-launch requirements being satisfied:

1. Purchase approval and security hardening of first 5 ruggedized industrial edge gateway devices

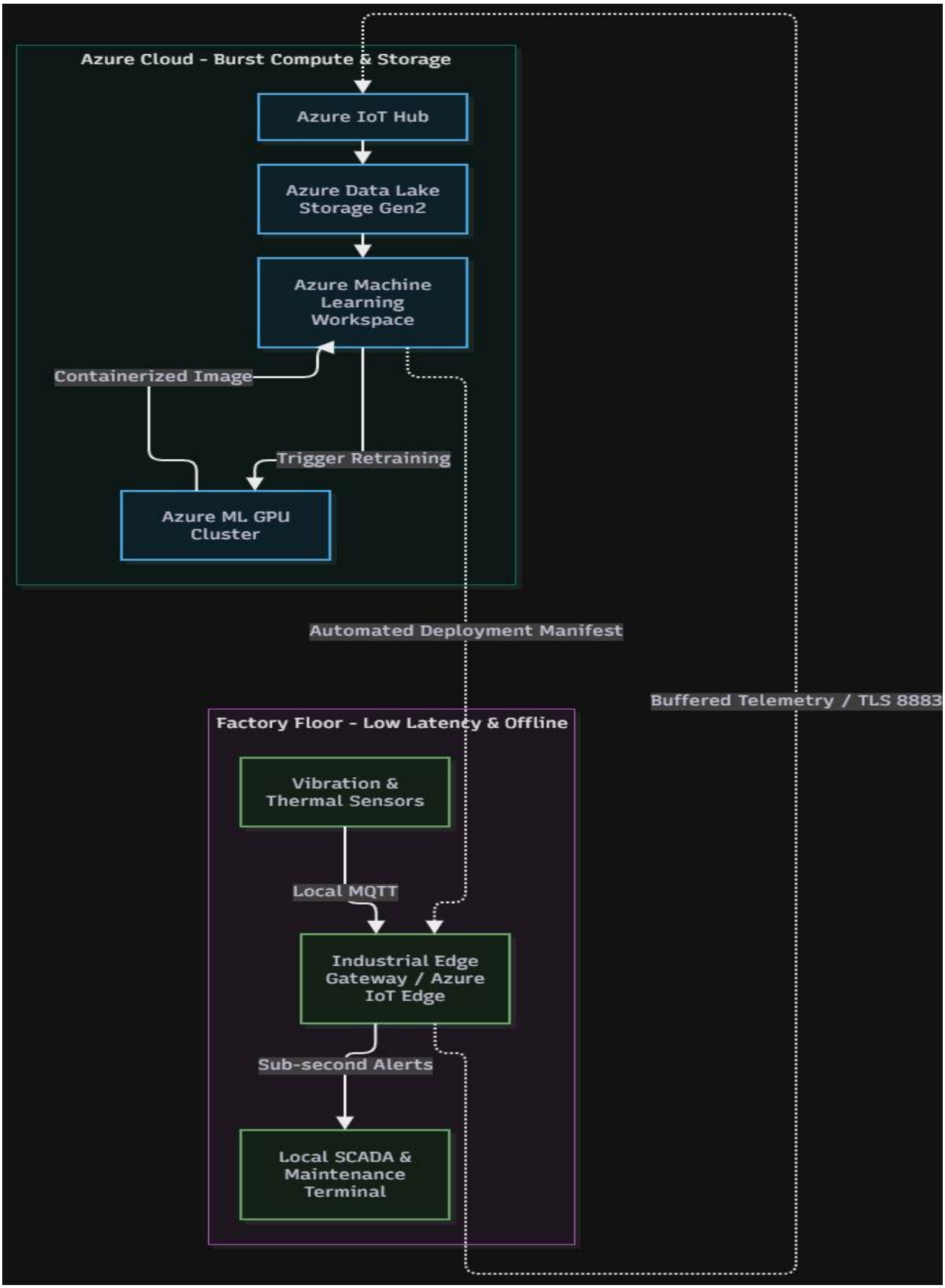
2. Azure Landing Zone network security assessment confirming IoT Hub egress (ports 8883/443) successfully traverses plant firewall without requiring ingress rules

3. Model achieves more than 88% precision and more than 92% recall on historical failure validation set during offline verification

5. Communication and Professional Practice

5.1 Deployment Architecture Diagram

The architecture below depicts the physical and logical boundary between the on-premise factory floor and Microsoft Azure cloud ecosystem.


Azure Cloud - Burst Compute & Storage
Azure IoT Hub
Azure Data Lake Storage Gen2
Azure Machine Learning Workspace
Containerized Image
Trigger Retraining
Azure ML GPU Cluster
Automated Deployment Manifest
Buffered Telemetry / TLS 8883
Factory Floor - Low Latency & Offline
Vibration & Thermal Sensors
Local MQTT
Industrial Edge Gateway / Azure IoT Edge
Sub-second Alerts
Local SCADA & Maintenance Terminal

5.2 References & Citations

Beyer, M. A., & Gartner, Inc. (2020). *Gartner glossary: Cloud computing*. Gartner. https://www.gartner.com/en/information-technology/glossary/cloud-computing

Huyen, C. (2022). *Designing machine learning systems: An iterative process for production-ready applications*. O'Reilly Media.

Mell, P., & Grance, T. (2011). *The NIST definition of cloud computing* (Special Publication 800-145). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-145

Microsoft. (2026). *Azure IoT Edge documentation: Deploy containers and operate offline*. Microsoft Learn. https://learn.microsoft.com/en-us/azure/iot-edge/

Microsoft. (2026). *Azure machine learning pricing and architecture*. Microsoft Azure. https://azure.microsoft.com/en-us/pricing/details/machine-learning/

Appendix A: Completed Readiness Scorecard

Completed evaluation template linking Week 2 readiness dimensions directly to deployment implications:

Dimension	Score (0-5) & Gap Type	Evidence & Effect on Deployment
Data	**Score:** 3 **Gap**: Monitor after launch	**Evidence:** Telemetry exists across machinery but labeling of failure events varies significantly across plant shifts. **Effect:** Requires cloud data lake ingestion to centralize and standardize ground truth
Model	**Score:** 4 **Gap:** Monitor after launch	**Evidence:** Supervised PoC algorithms achieve 89% accuracy on historical logs but have not been stress tested against operational drift **Effect:** Requires automated ML retraining workflow in the cloud
Infrastructure	**Score:** 2 **Gap:** Fix before launch	**Evidence:** Plant floor has legacy PLCs and no inference-grade compute units **Effect:** Forces purchase of industrial edge gateway hardware before deployment
Organizational	**Score:** 3 **Gap:** Fix before launch	**Evidence:** Plant has mechanical technicians but no dedicated MLOps/cloud engineers **Effect:** Skews preference toward managed PaaS solutions (Azure ML) vs custom bare metal stacks
Risk	**Score:** 4 **Gap:** Monitor after launch	**Evidence:** False alarms result in inspection downtime but missed failures trigger total production shutdowns

		Effect: System designed as advisor rather than autonomous, requires human-in-the-loop confirmation before shutdown
Cost	**Score:** 3 **Gap:** Fix before launch	**Evidence:** Cloud operations budget exists but requires approval for upfront hardware purchases **Effect:** Incremental deployment needed to demonstrate ROI before full fleet hardware buy

Appendix B: Environment Fit Scorecard & Comparison Templates

Table B1: Environment Fit Scorecard Summary

Question (in order)	Facts About My Use Case	Score (1-5)
1. Data sensitivity	Proprietary industrial telemetry; sensitive trade secret data but free from consumer PII/HIPAA restrictions.	3 (Balanced)
2. Cost pattern	Continuous basic monitoring, but heavy burst computing needed for model retraining.	3 (Pulls Cloud)
3. Operational capacity	Small plant IT team that cannot manage physical GPU server farms.	5 (Pulls Cloud)
4. Latency	Immediate action required on factory floor; network drops are common.	1 (Pulls On-Premise)

Average: 3.0, Spread: 4, Data/Latency scored 1? Yes (Latency = 1), Decision Rule: Hybrid Strategy

Table B2: Three-Environment Comparison for Use Case

Dimension	On-Premise	Cloud	Hybrid
Data control	Full physical sovereignty, data never leaves plant network.	Requires TLS transit or strict cloud compliant encryption.	Inference data stays local, aggregated telemetry shipped to cloud.
Cost model	High upfront Capital expenditure for edge hardware and GPU	Pure Operational expenditure, pay-as-you-go for storage	Capital expenditure for edge devices, Operational

	servers.	and burst compute.	expenditure for cloud training and ingestion.
Scalability	Restricted by physical on-site rack capacity.	Almost unlimited compute for big data retraining	Cloud training infinitely scalable, edge scales per machine..
Operational burden	Extremely high, factory IT handles all hardware and OS patching.	Low, Microsoft manages infrastructure/clusters	Moderate, IT manages edge hardware health while cloud is managed.
Best fit for system?	No, unjustified Capital expenditure for idle retraining GPUs.	No, internet outages interrupt critical machine safety alerts.	Yes, addresses offline latency while preserving cloud scalability.

Appendix C: Service Map and Cost Templates

Table C1: Service Map (Platform: Microsoft Azure)

Building Block	Named Service	Service Model	Container / Serverless	Why
Training compute	Azure Machine Learning Compute	PaaS	Serverless	Auto-scales GPU clusters on demand; shuts down when idle.
Serving/hosting	Azure IoT Edge	PaaS / Edge	Container	Enables Dockerized ML inference offline on factory gateways.
Data storage	Azure Data Lake Storage Gen2	PaaS	Serverless	High-throughput, cost-effective storage for raw telemetry.
Managed AI/ML	Azure Machine Learning Workspace	PaaS	Serverless	Provides model registry, tracking, and edge deployment manifests.
Additional: Ingestion	Azure IoT Hub	PaaS	Serverless	Secure bidirectional telemetry and edge device fleet management.

Table C2: Three-Year Cost Comparison (Order of Magnitude)

Cost Line	Assumptions & Source (Accessed Oct 2026)	Option A: Hybrid	Option B: Full On-Prem	Uncertainty (H/M/L)
Compute	50 edge nodes ($500/ea); 50 hrs/mo Azure NC-series GPU vs. on-prem GPU server	$25,000 edge CapEx + $7,920 cloud GPU	$25,000 edge CapEx + $42,000 server Capital Expediture	Medium
Storage & Transfer	5 TB/mo raw telemetry; 200 GB/mo compressed uploaded to Data Lake	$4,320 (Blob storage + egress)	$10,800 (Local SAN disks + backup)	Low
Managed Services	Azure IoT Hub Standard tier (S2) + Azure ML Workspace	$10,800 (IoT Hub & registry)	$0 (Open-source tools)	Low
Staff Time to Operate	Plant IT admin time to maintain infrastructure and deployments	0.5 FTE (~$120,000 / 3 yrs)	1.0 FTE (~$240,000 / 3 yrs)	High
Three-Year Total	Comprehensive infrastructure + operational labor footprint	~$168,040 (Total)	~$317,800 (Total)	Medium